

ARSHAD

Backend / SDE Intern Candidate

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PROFESSIONAL SUMMARY

Final-year Backend Development student (TISS SSE) seeking an SDE/Backend Engineering internship. Shipped three production systems end-to-end (Spring Boot/WebSockets, PHP/MariaDB, Spring Boot/MongoDB Atlas) and built a GenAI/RAG pipeline (LangChain, HuggingFace, Gemini). Strong in REST API design, event-driven backend logic, and relational + NoSQL databases.

TECHNICAL SKILLS

Languages: Java, Python, PHP, JavaScript, C, HTML5, CSS3

Backend & APIs: Spring Boot, Spring Security, Spring MVC, RESTful API Design, Native WebSockets, Event-Driven Architecture

Databases: MongoDB (Atlas), MySQL, MariaDB, PostgreSQL/pgvector, FAISS

GenAI/RAG: LangChain, HuggingFace, Google Gemini API, FAISS, MongoDB Atlas Vector Search

Reliability & Security: Bean Validation, CORS Configuration, IP-Based Rate Limiting, Global Exception Handling, Health Checks

Testing & Tools: JUnit, Postman, Git/GitHub, GitHub CLI, Docker, Maven, Gradle, Ubuntu Linux (CLI)

SELECTED PROJECTS

Real-Time Enterprise Messaging Engine (*Deployed*) — Java, Spring Boot, WebSockets, MongoDB, Docker

- Migrated from STOMP to native WebSockets; built a thread-safe ConcurrentHashMap routing layer for O(1) average-case connection lookup
- Fixed a WebSocket identity-spoofing vulnerability by binding session auth at the handshake, and root-caused a TOCTOU race condition in concurrent message handling
- Designed a single-collection MongoDB schema supporting direct and group chats; shipped via Docker/CI-CD on Render

Hospital Management System (*Deployed*) — PHP, MariaDB, HTML5, CSS3, JavaScript

- Fixed a privilege-escalation vulnerability, broken JOIN queries, and a broken bcrypt password-hashing flow in a live MVC-restructured system
- Recovered and rebuilt 22 missing view files by tracing Git history, restoring core functionality before redeploying on InfinityFree

Full-Stack Developer Portfolio (*Deployed*) — Java, Spring Boot, MongoDB Atlas, JavaScript

- Scaffolded a layered Spring Boot backend (Controller/Service/Repository/DTO); diagnosed and fixed a MongoDB Atlas connection failure caused by a malformed URI
- Configured origin-restricted CORS and IP-based rate limiting; validated via PowerShell API testing and browser checks

End-to-End RAG Pipeline (*Independent*) — Python, LangChain, HuggingFace, Google Gemini API, FAISS

- Built a full retrieval-augmented generation pipeline covering ingestion, chunking, embedding, vector storage, and generation
- Used HuggingFace models for embeddings and Gemini API for generation, orchestrated via LangChain; implemented FAISS as a self-hosted local vector store for similarity search

WORK EXPERIENCE

Software Development Intern — PC Training Institute (PCTIL) Aug 2025 – Oct 2025

- Built billing and appointment-scheduling modules of a hospital management backend in PHP/MySQL within an existing multi-module codebase
- Optimized SQL queries and added indexing on frequently-filtered columns, speeding up appointment and billing lookups

EDUCATION

Bachelor of Software Development — Tata Institute of Social Sciences (TISS SSE)

Mumbai, India | Expected 2027 | GPA: 8.4/10 | Relevant coursework: DSA, OOP, DBMS, Software Engineering

CERTIFICATIONS

- RAG with MongoDB — MongoDB via Credly (Aug 2026)
- Software Development Job Simulation — Citi ICG Technology, via Forage (Apr 2026)